NET (BMEVITMAC09): one exercise per theme, sessions 1 to 6

Written by us from the 2026 slides (hta1 to hta6p), pages marked "Not exam material" excluded. Goal: be able to answer on every theme of the midterm. Each theme has one short exercise with an empty box; the second part repeats the exercises with the boxes filled in. The exercises complement the three mock papers (`examens-blancs-net.pdf`) and do not repeat them; the coverage table points to both. Calculations are checked by `calc_exos_par_theme_net.py`.

Coverage table

Session 1 (hta1, hta1p)

Ex.	Theme (slides)	Mock papers
1.1	Telephone signal, PCM	
1.2	Telecommunication history, PSTN	
1.3	Multiplexing: FDM, TDM, CDM	
1.4	Multiplexing vs multiple access, FDMA, TDMA, CDMA	A6(a)
1.5	Duplexing: FDD, TDD, wires	A6(b)
1.6	Channel arbitration	
1.7	Communication patterns and addressing	
1.8	Assigning addresses	
1.9	Centralized vs distributed models	
1.10	Protocol, stack, SAP, PDU	
1.11	OSI and TCP/IP models	C1(a)
1.12	Encapsulation, layering violation	C1(b)
1.13	Standardization, conformance, fuzzing	
1.14	Security: encryption, authentication, integrity, AAA	
1.15	QoS: IntServ, DiffServ, CAC, UNI	B6(d)
1.16	Token bucket	A6(c)
1.17	Leaky bucket, RED	A6(c)
1.18	Statistical multiplexing	C1(c)
1.19	Serialization, byte order	
1.20	TLV	B5(c)
1.21	ASN.1 (p. 22)	
1.22	XML tools (DTD, XPath, XSLT…)	
1.23	XML, JSON, YAML conversion	A5, B5
none	Mobile telephony (p. 14): out of scope per the slides	
none	Standardization bodies, ASN.1 example (hta1p p. 9 to 11, 23): not exam material	
none	Erlang model (hta1p p. 20): only cited	

Session 2 (hta2, hta2p)

Ex.	Theme (slides)	Mock papers
2.1	Network layer tasks, circuit vs packet switching	
2.2	IPv4 service, hourglass model	
2.3	Direct delivery, gateway, addressing modes	
2.4	Hop-by-hop vs source routing	
2.5	Routing loops, TTL	C2(d)
2.6	Fragmentation, MTU	
2.7	IPv4 header fields	
2.8	IHL, options, checksum, DSCP, ECN	
2.9	Classful addressing	
2.10	Problems of classes, subnetting	
2.11	CIDR notation, network sizes	A1(a), B1, C2(a)
2.12	Special and private addresses	C2(d)
2.13	CIDR allocation strategy	
2.14	Mobile IP	
2.15	Same subnet	A1(c)
2.16	Longest prefix match	A2
2.17	Routing table aggregation	B2, C2(d)
none	IGMP and multicast groups (hta2 p. 6): only cited	
none	IPv6 (hta2 p. 17, hta2p p. 9): only cited	

Session 3 (hta3, hta3p)

Ex.	Theme (slides)	Mock papers
3.1	Data, control, management planes	
3.2	Signalling, in-band and out-of-band	
3.3	ICMP messages, ping, traceroute	C3(c)
3.4	ICMP security	
3.5	DHCP options, lease, dhclient, Kea	C3(a)
3.6	DORA with several servers	C3(a)
3.7	URL and URI	
3.8	TLDs, FQDN	
3.9	DNS records, resolvers, lookup	C3(b), C6(a)
3.10	Domain ownership, ICANN, WHOIS	
3.11	DNS transport and security	
3.12	DNS tools and servers	
3.13	ifconfig, route, netstat, iproute2, ipconfig	
3.14	Permanent settings, management utilities	
3.15	Ping and traceroute probes	C3(c)
3.16	tcpdump, Wireshark, tshark	
3.17	Mininet	
none	ARP (hta3 p. 2): only cited	
none	Windows, Linux distributions, desktops, shell basics (hta3p p. 3 to 9): environment only	

Session 4 (hta4, hta4p)

Ex.	Theme (slides)	Mock papers
4.1	Routing, forwarding, RIB, FIB	
4.2	Router capacity	
4.3	Optimal path, 1:1 and 1+1 redundancy	
4.4	Flat and hierarchical routing	
4.5	Distance vector convergence	C4
4.6	RIP limit, count to infinity	B3
4.7	Link state database, Dijkstra	A3
4.8	Algorithm complexity	
4.9	RIP versions	
4.10	OSPF	
4.11	IS-IS	
4.12	Autonomous systems, IGP, EGP	C4(c)
4.13	Peering and transit	
4.14	BGP	C4(c)
4.15	Implementations, BIRD demo	
none	Router examples (hta4 p. 4): product links only	
none	Sensor networks (hta4 p. 17): not exam material	

Session 5 (hta5, hta5p)

Ex.	Theme (slides)	Mock papers
5.1	Transport and session in the IP model	
5.2	Identifiers: MAC, IP, port	
5.3	Port allocation, 5-tuple	B4(b)
5.4	Flow vs congestion control	
5.5	Datagrams and streams	
5.6	UDP	C1(b)
5.7	TCP features, DCCP, SCTP, QUIC over UDP	
5.8	Connection setup and teardown, SYN flood	C5(a)
5.9	Acknowledgement, sliding window	C5(b)
5.10	TCP header and options	C5(c)
5.11	NAT	B4
5.12	Flow control, Silly Window Syndrome	
5.13	Bottleneck	
5.14	Congestion collapse, goodput	
5.15	cwnd, adv_wnd, effective window	
5.16	Slow start, short connections	A4
5.17	Tahoe, Reno, fast retransmit, fast recovery	A4, C5(b)
5.18	Long Fat Network, window scaling	C5(c)
5.19	Links with high loss rate	
5.20	TCP variants, CUBIC	
5.21	What programmers and operators should know	
none	Connection hijacking, reset (hta5 p. 10): only cited	
none	Jokes (hta5 p. 17): not exam material	

Session 6 (hta6, hta6p)

Ex.	Theme (slides)	Mock papers
6.1	World Wide Web, search, W3C	
6.2	Browsers and engines	
6.3	Web servers and proxies	
6.4	HTML, XHTML, CSS	
6.5	JavaScript, DOM, AJAX	
6.6	HTTP methods and response codes	
6.7	User-Agent and Referrer	
6.8	HTTP versions	B6(a)
6.9	HTTP state, cookies	
6.10	Symmetric and asymmetric cryptography	
6.11	Publishing public keys, PKI	B6(b)
6.12	SSL and TLS	B6(a)
6.13	QUIC	B6(c)
6.14	E-mail origins	
6.15	E-mail agents	C6(a)
6.16	Message format, MIME	C6(c)
6.17	Mail storage	
6.18	SMTP	C6(a)
6.19	POP3 and IMAP	C6(b)
6.20	Mail software, webmail, Exchange	
6.21	Client authentication	
6.22	OpenID, OAuth, TOTP	
6.23	PGP, S/MIME	
6.24	Fighting spam, DKIM, SPF, DMARC	C6(d)
6.25	Phishing	
none	WWW alternatives (hta6 p. 16): not exam material	
none	Origin of the word spam (hta6p p. 16 and 17): no network content	

Exercises

Session 1: introduction and communication protocols

1.1 Telephone signal and PCM hta1 p. 13

Speech on a telephone line is limited to 0.3 to 3.4 kHz. Why is it sampled at 8 kHz? What is the bitrate of one PCM channel with 8-bit samples? How many such channels fit in 2.048 Mbit/s if no capacity is spent on anything else? Why are the quantization steps logarithmic?

1.2 Telecommunication history hta1 p. 5 to 12

Why do the most common English letters have the shortest Morse codes? What does a telephone exchange do, and how is the PSTN organised?

1.3 Multiplexing: FDM, TDM, CDM hta1 p. 16 and 17

Two streams share a channel with CDM. Codes $c_1 = (+1, +1)$, $c_2 = (+1, -1)$; bit 1 is sent as $+1$, bit 0 as -1; the channel carries $b_1c_1 + b_2c_2$. Source 1 sends 1, source 2 sends 0. Give the signal and decode both bits by multiplying with each code, summing and dividing by 2. Which of FDM, TDM, CDM can analog signals not use?

1.4 Multiplexing or multiple access hta1 p. 16 to 19

Classify the downlink of a mobile tower and the uplink of the phones. Who assigns the channels in multiplexing? Why can CDMA codes not be fully orthogonal in multiple access?

1.5 Duplexing hta1 p. 20

A 20 Mbit/s radio link uses TDD and gives 3/4 of the time to the downlink. Throughput in each direction? How do Ethernet or USB cables do duplexing? With FDD, what fixes the capacity of each direction?

1.6 Channel arbitration hta1 p. 21

Give the slide example for each method: fixed allocation, central entity on request, contention. Which method has collisions?

1.7 Communication patterns and addressing hta1 p. 22 and 23

For unicast, broadcast, multicast and geocast, what kind of address does the communication need?

1.8 Assigning addresses hta1 p. 24; hta2 p. 6

Give three ways to assign addresses. In IPv4, what does an address identify? At least how many addresses does a router with three interfaces have?

1.9 Centralized and distributed models hta1 p. 25

Compare the two models and give one example pair from the slides.

1.10 Protocol, stack, SAP, PDU hta1p p. 2 and 3

Define a communication protocol. What is a Service Access Point (give example methods)? With which layers does a layer talk directly, and with which one virtually?

1.11 OSI and TCP/IP models hta1p p. 5 and 6

Why is OSI not used in practice? What does TCP/IP define instead of layer responsibilities? Which OSI layer handles character encoding, compression and encryption?

1.12 Encapsulation hta1p p. 4; hta2 p. 10; hta5 p. 13

1000 octets of application data are sent over TCP and IPv4, both without options. What is the IPv4 Total Length? What may a layer add besides a header?

1.13 Standardization and testing hta1p p. 7 and 8

In the black box model, what is rigorously specified and what is not? What is a conformance test, and fuzz testing?

1.14 Security mechanisms hta1p p. 12 to 15

What did Diffie-Hellman solve? What does challenge-response protect against? Why is a simple checksum not enough for integrity? What does AAA stand for?

1.15 QoS: reservation, admission, UNI hta1p p. 16 and 17

What does a client register when negotiating QoS limits? Who accepts or refuses a request? Where is excess traffic dropped, and where are flows identified?

1.16 Token bucket hta1p p. 18

A token bucket has $K = 2$ Mbit/s and $B = 1$ Mbit (one token per bit), and is full. The client sends at 6 Mbit/s. How long is the flow unaffected, and how much data passes in that time?

1.17 Leaky bucket and RED

hta1p p. 19

A leaky bucket outputs at most $K = 2$ Mbit/s and can hold $B = 1$ Mbit. Input: 6 Mbit/s for 0.25 s, then nothing. Is anything dropped, and when has everything left? What does RED do?

1.18 Statistical multiplexing

hta1p p. 20

$N = 200$ users, each active with $p = 5\%$. Give the mean, the binomial variance $Np(1-p)$ and σ. With Chebyshev and $k = 2$, which capacity L bounds the blocking probability by 1/4?

1.19 Serialization: byte order

hta1p p. 21

Port 8080 is stored on 16 bits. Give the two octets in network order and in memory on an Intel host. Which function converts it? Why avoid floats?

1.20 TLV

hta1p p. 21 and 22

Type and Length take one octet each; type 1 = name (ASCII), type 2 = port (16 bits, big endian); the reader knows no other type. Decode `01 03 77 65 62 09 02 AA BB 02 02 1F 90`.

1.21 ASN.1

hta1p p. 22

What is ASN.1, how are its messages serialized, and why do X.509 certificates use DER? Name two other users.

1.22 XML tools

hta1p p. 24

What are DTD and XSD, XPath, XSLT, namespaces and SAX used for?

1.23 YAML to JSON and XML

hta1p p. 24 to 26

Convert to JSON and to XML (root `<config>`, list items `<port>`). Then give two YAML problems from the course.

```
service: web
ports:
  - 80
  - 443
admin:
  name: Kata
  mail: kata@example.org
```

Session 2: IPv4

2.1 Network layer tasks

hta2 p. 2

Give the three tasks of the network layer and the difference between circuit and packet switching.

2.2 IPv4 service and hourglass model

hta2 p. 3, 4 and 17

What does IPv4 expect from the link layer and offer upwards? In the hourglass model, which layers are private to neighbours, private to endpoints, shared by everyone? Why is IPv4 hard to replace?

2.3 Direct delivery, gateway, addressing modes

hta2 p. 5 and 6

Host 10.1.2.3/24 with gateway 10.1.2.1 sends to 10.1.2.77 and to 10.1.3.5. Direct or through the gateway? Can its broadcast reach 10.1.3.0/24?

2.4 Hop-by-hop and source routing

hta2 p. 7; hta4 p. 6

Compare both, with one advantage and one drawback each, and where source routing is used.

2.5 Routing loops and TTL

hta2 p. 8; hta3 p. 4

A packet leaves with TTL 12, crosses R1, R2, R3, then enters a loop R4, R5, R4… Which router drops it, and how many times does it cross the R4-R5 link? Why don't routers just remember the packets they have seen?

2.6 Fragmentation

hta2 p. 9 and 12

A packet with Total Length 2400 (20-octet header) must cross a link with MTU 1006. Give data size, Total Length, Fragment Offset and MF of each fragment. Does IPv4 do path MTU discovery?

2.7 IPv4 header fields

hta2 p. 10 to 15

Decode the first 10 octets of a header: `45 00 05 DC 1C 46 40 00 40 11`.

2.8 IHL, options, checksum, DSCP, ECN

hta2 p. 10, 11, 14, 16

IHL is 7: header length and options length? Why are options rarely used? Why must every router recompute the Header Checksum? What are DSCP and ECN for?

2.9 Classful addressing hta2p p. 3 and 4

Give the class and say whether it could be assigned to a host before CIDR: 10.200.0.1, 127.0.0.1, 130.50.255.255, 200.1.1.0, 200.1.1.1, 230.1.2.3, 245.0.0.1.

2.10 Problems of classes, subnetting hta2p p. 5

An organisation has 2000 computers. How many class C blocks does it need, and which single CIDR prefix length would do? What is the three-level hierarchy of subnetting?

2.11 CIDR notation hta2p p. 6, 7 and 11

Convert 255.255.252.0 to a prefix length and /13 to a netmask. Is 255.255.0.255 valid? How many hosts in a /19?

2.12 Special and private addresses hta2p p. 4 and 8

Which are private, loopback or neither: 172.15.3.1, 172.31.255.1, 192.169.1.1, 10.255.0.1, 127.0.0.53?

2.13 CIDR allocation strategy hta2p p. 9

How should addresses be allocated for manageable routing tables, and why are IPv4 tables “destined to be a mess forever”?

2.14 Mobile IP hta2p p. 10

Why does moving a device to another network break its connections, and what does Mobile IP do?

2.15 Same subnet, dotted mask hta2p p. 12

Are 10.20.30.40 and 10.20.33.1 on the same subnet with mask 255.255.248.0?

2.16 Longest prefix match hta2p p. 13

Table: 172.16.0.0/12 if0; 172.20.0.0/14 if1; 172.22.128.0/17 if2; 0.0.0.0/0 if3. Interfaces for 172.23.200.1, 172.22.130.5, 172.31.0.1, 172.32.0.1?

2.17 Routing table aggregation hta2p p. 14

The lines 192.0.4.0/24 to 192.0.8.0/24 (five /24) all go to eth1. Merge them.

Session 3: IP protocols and tools

3.1 Data, control, management planes hta3 p. 2

Place HTTP, RTCP, SNMP, ICMP and OSPF in the planes of the slide. How has the definition of routing changed? Where are ARP and DHCP?

3.2 Signalling hta3 p. 3 and 4

In-band or out-of-band? Why is ICMP in-band for IP but out-of-band for TCP and UDP? How do ITU-T and IETF differ?

3.3 ICMP messages hta3 p. 4 to 6

Which ICMP type for: TTL reaches 0; datagram to a closed port; DF set but MTU too small; a router knows a better route; ping?

3.4 ICMP security hta3 p. 7

Give three abuses of ICMP and what firewalls do.

3.5 DHCP server, options, tools hta3 p. 8 to 10; hta3p p. 15

A server hands out 192.168.50.100 to 192.168.50.149 with 1 hour leases. How many clients at most? How are options encoded, and which are typical? Name the usual client and server.

3.6 Several DHCP servers hta3 p. 10

Two DHCP servers are active on one LAN. What happens during DORA, and why is Request broadcast?

3.7 URL and URI hta3 p. 12

Split `https://docs.example.org/net/tcp.html`. Which part does DNS resolve? URL vs URI?

3.8 TLDs and FQDN hta3 p. 13

Classify .hu, .org, .gov and .co.uk. What is an FQDN?

3.9 DNS records and lookup hta3 p. 14 and 15

Which record does a mail server need to find the MTA of example.net? Who holds a subtree? What is a reverse lookup?

3.10 Domain name ownership hta3 p. 16

Who manages TLDs and root servers, who runs .org, how do you rent a name and check its owner?

3.11 DNS transport and security hta3 p. 17

When is TCP used instead of UDP? Name four weaknesses. Why does DNS over HTTPS bypass the original mechanism?

3.12 DNS tools and servers hta3 p. 18

Name three query tools and two servers.

3.13 Configuration commands hta3p p. 4, 10 to 14

With iproute2: give eth0 address 172.16.5.20/24, bring it up, add a route to 10.40.0.0/16 via 172.16.5.1. Old equivalents? What does `netstat -pnaut` list? Windows equivalent of ifconfig?

3.14 Permanent settings hta3p p. 16 and 17

Write `/etc/network/interfaces` for eth0 by DHCP and eth1 static 172.16.5.20/24, gateway 172.16.5.1. Where is the DNS server configured? Which tool has its own DHCP client?

3.15 Ping and traceroute hta3p p. 18 to 20

Can ping succeed while web traffic fails? Which probes can traceroute send?

3.16 Capture tools hta3p p. 21 to 24

What do tcpdump, Wireshark and tshark do? What is promiscuous mode?

3.17 Mininet hta3p p. 25; hta4p p. 15

What is Mininet and what does its Python API do?

Session 4: routing

4.1 Routing, forwarding, RIB, FIB hta4 p. 2 and 3; hta4p p. 2

Define forwarding and routing. What can the RIB contain that the FIB does not? What makes a Linux machine a router?

4.2 Router capacity hta4 p. 3

A 10 Gbit/s interface: packets per second with 1500-octet packets, and with 64-octet packets?

4.3 Optimal path and redundancy hta4 p. 5

List four optimality criteria. With 1+1 redundancy over two disjoint paths each losing a packet with probability 1% independently, what is the probability of losing both copies? Why disjoint?

4.4 Flat and hierarchical routing hta4 p. 6

Compare flat and hierarchical models.

4.5 Distance vector convergence hta4 p. 7; hta4p p. 4 and 5

8 routers form a ring (hop count). After how many rounds does every router know every destination? What does an advertisement contain?

4.6 RIP hop limit hta4 p. 10; hta4p p. 6

A RIP network has a diameter of 17 hops. Consequence? Why weren't more fields added to the advertisements to fix count to infinity?

4.7 Link state database hta4p p. 7 and 8

Links W-X 3, W-Y 1, X-Y 1, X-Z 4, Y-Z 6. Write the link state database and the shortest distances from W. Give the four steps of link state routing.

4.8 Algorithm complexity hta4p p. 3

$|V| = 1000$, $|E| = 5000$: compare $|E| + |V| \log |V|$ (base 2) with $|V|^2$. Which algorithm for LS, which for DV, and why?

4.9 RIP versions hta4 p. 10

What changed in RIPv2 and RIPng? Metric and update interval of RIP?

4.10 OSPF hta4 p. 11

What are areas and the backbone? Is 0.0.0.0 an address here? Why a Designated Router? Metric?

4.11 IS-IS hta4 p. 12

Give three differences with OSPF.

4.12 Autonomous systems hta4 p. 13

What is an AS? Pros and cons? IGP and EGP examples?

4.13 Peering and transit hta4 p. 14

ISP1 peers with ISP2, ISP2 peers with ISP3. Can ISP1 reach ISP3 through ISP2?

4.14 BGP hta4 p. 15

Number of iBGP sessions for a full mesh of 12 routers? With 3 route reflectors in full mesh, each serving 3 clients that peer only with their reflector? Why is BGP routing "policy routing"?

4.15 Implementations and demo hta4 p. 16; hta4p p. 15 to 17

What does a routing daemon do? Name two open-source ones. In the demo, why does a ping to an address on a non-adjacent link fail before BIRD starts?

Session 5: transport layer and TCP

5.1 Transport and session in the IP model hta5 p. 2

Who handles keepalive and multiplexing in TCP/IP compared with OSI?

5.2 Identifiers hta5 p. 3

Size and scope of MAC address, IP address and port? How many port numbers?

5.3 Port allocation and 5-tuple hta5 p. 4

A browser opens two connections to the same server on port 443. What distinguishes them? Which ports are well-known? Which protocol fixes both ports?

5.4 Flow and congestion control hta5 p. 5

Who is protected by each, and how is congestion detected?

5.5 Datagrams and streams hta5 p. 6; hta5p p. 20

An application writes three 100-octet messages to a TCP socket. Will the receiver read three 100-octet messages? And with UDP?

5.6 UDP hta5 p. 7

UDP carries 100 octets: Length field and IPv4 Total Length? What does UDP add to IP?

5.7 TCP, UDP and in between hta5 p. 8, 20 to 23

Place TCP, UDP, DCCP and SCTP. Why are QUIC and μTP built over UDP?

5.8 Connection start and end hta5 p. 9 and 10

Who may close a connection, with which segments? Why are initial sequence numbers random? Other attacks besides SYN flood?

5.9 Acknowledgement and sliding window hta5 p. 11; hta5p p. 2

Window of 4 segments; ACK 3 has arrived. Which segments may be sent? And after ACK 5?

5.10 TCP header hta5 p. 13 to 16

Data Offset is 8: header and options length? What do PSH and URG mean? Largest window with window scaling?

5.11 NAT trade-offs hta5 p. 18 and 19

A gateway can use 28,000 ephemeral ports: how many simultaneous connections to one server port? Two benefits and two problems of NAT?

5.12 Flow control, Silly Window Syndrome hta5p p. 3

What happens when adv_wnd = 0? Causes and fixes of the Silly Window Syndrome?

5.13 Bottleneck hta5p p. 4

A path has links of 1 Gbit/s, 100 Mbit/s, 20 Mbit/s, 1 Gbit/s. Bottleneck? Why do bottlenecks move?

5.14 Congestion collapse and goodput hta5p p. 5 and 6

Load 100 Mbit/s, of which 30 Mbit/s are duplicates or lost. Goodput? Where should the load stay? Why not reserve bandwidth for every flow?

5.15 Effective window hta5p p. 7

MSS 1460 octets, cwnd 10 MSS, adv_wnd 8760 octets. Effective window?

5.16 Slow start and short connections hta5p p. 8

cwnd starts at 1, ssthresh is large, no loss. How many round trips to send 60 segments? Why does HTTP suffer?

5.17 Tahoe and Reno hta5p p. 9 to 15

cwnd 1, ssthresh 8, slow start capped at ssthresh, CA +1; a loss is detected at the end of round 8. Give cwnd for rounds 1 to 12 with Tahoe (timeout) and Reno (3 duplicate ACKs), and the segments sent in rounds 9 to 12.

5.18 Long Fat Network

hta5p p. 16

Satellite link: 10 Mbit/s, RTT 600 ms. Bandwidth-delay product, throughput without window scaling, minimum shift?

5.19 Links with high loss rate

hta5p p. 17

Why does Reno perform badly on Wi-Fi, and which solutions do the slides give?

5.20 TCP variants

hta5p p. 18 and 19

Match: New Reno, Vegas, Compound, CUBIC. Which variant is the default today?

5.21 Programmers and operators

hta5p p. 20 and 21

Why reuse connections? What is `SO_REUSEPORT` for? Why is TCP hard for operators?

Session 6: Web and e-mail

6.1 World Wide Web

hta6 p. 2 and 3

What is hypertext? Directory vs search engine? What does robots.txt do? Who standardizes?

6.2 Browsers

hta6 p. 4

What is a browser engine? Which engines are still developed, and why is Blink's dominance a problem?

6.3 Servers and proxies

hta6 p. 5

Name two web servers and four uses of a web proxy.

6.4 HTML

hta6 p. 6

What is XHTML, what superseded it, and what describes formatting separately?

6.5 JavaScript

hta6 p. 7

What are the DOM and AJAX?

6.6 HTTP methods and response codes

hta6 p. 8

Which method to fetch a page, to send a search term, to get only metadata? Meaning of 200, 301, 403, 404, 500, 502, 504?

6.7 User-Agent and Referrer

hta6 p. 9

Why do browsers lie in the User-Agent string? What is the Referrer for?

6.8 HTTP versions

hta6 p. 10

A page has 1 HTML file and 19 images on one server. TCP handshakes with one connection per object vs HTTP/1.1 reuse? What do HTTP/2 and HTTP/3 add?

6.9 State and cookies

hta6 p. 11

Why can't a response with a cookie be cached, and the workaround? What is a third-party cookie?

6.10 Symmetric and asymmetric cryptography

hta6 p. 12

What does Diffie-Hellman not protect against? Encrypt for Bob, sign as Alice: which keys? Why sign a hash?

6.11 Publishing public keys

hta6 p. 13

Web of trust vs PKI. Three criticisms of PKI?

6.12 SSL and TLS

hta6 p. 14

Relation between SSL and TLS, deprecated versions, layer, StartTLS?

6.13 QUIC drawbacks

hta6 p. 15

Give three limits of QUIC from the slides.

6.14 E-mail origins

hta6p p. 3

What did `username@computer` mean originally, and now?

6.15 Agents

hta6p p. 4

Role of MTA, MDA, MUA, Message Submission Agent, Message Retrieval Agent?

6.16 Message format

hta6p p. 5

Write the header of a message from Anna Kis to two recipients, then where the body starts. Give a MIME type.

6.17 Mail storage

hta6p p. 6

Compare mbox and maildir.

6.18 SMTP

hta6p p. 7

Command sequence, transport, ports, response codes?

6.19 POP3 and IMAP commands

hta6p p. 8 and 9

Name the commands and how IMAP handles the Sent folder.

6.20 Mail software

hta6p p. 10 to 12

How is webmail usually implemented? Name an MTA. What is special about Exchange?

6.21 Client authentication

hta6p p. 13

Three factors? When is PLAIN acceptable? How does CRAM work and what does it prevent?

6.22 OpenID, OAuth, TOTP

hta6p p. 14

What does each provide?

6.23 PGP and S/MIME

hta6p p. 15

What does PGP use instead of CAs? What is S/MIME for?

6.24 Fighting spam

hta6p p. 16 to 18

Two MTA acceptance policies and what MUA filters look at.

6.25 Phishing

hta6p p. 19

What is phishing and why is full protection impossible?

Solutions

Session 1: introduction and communication protocols

1.1 Telephone signal and PCM

hta1 p. 13

Speech on a telephone line is limited to 0.3 to 3.4 kHz. Why is it sampled at 8 kHz? What is the bitrate of one PCM channel with 8-bit samples? How many such channels fit in 2.048 Mbit/s if no capacity is spent on anything else? Why are the quantization steps logarithmic?

Nyquist: sample at twice the maximum frequency; speech stays intelligible below 4 kHz (filters keep a guard band), so 8 kHz. $8000 \times 8 = 64$ kbit/s. $2048/64 = 32$ channels. Logarithmic steps match human perception.

1.2 Telecommunication history

hta1 p. 5 to 12

Why do the most common English letters have the shortest Morse codes? What does a telephone exchange do, and how is the PSTN organised?

Frequent letters cost less, so messages are shorter on average. An exchange (switchboard) connects subscriber lines. The PSTN is hierarchical (local, regional, international exchanges), uses undersea cables and satellites between continents, and gives every endpoint a unique E.164 number.

1.3 Multiplexing: FDM, TDM, CDM

hta1 p. 16 and 17

Two streams share a channel with CDM. Codes $c_1 = (+1, +1)$, $c_2 = (+1, -1)$; bit 1 is sent as $+1$, bit 0 as -1; the channel carries $b_1 c_1 + b_2 c_2$. Source 1 sends 1, source 2 sends 0. Give the signal and decode both bits by multiplying with each code, summing and dividing by 2. Which of FDM, TDM, CDM can analog signals not use?

Signal $(+1, +1) + (-1, +1) = (0, 2)$. With c_1: $(0 + 2)/2 = +1$, bit 1. With c_2: $(0 - 2)/2 = -1$, bit 0. The codes are orthogonal ($1 - 1 = 0$), so the channels separate although they use the same band at the same time. TDM: the slides note that analog cannot take turns in time slots.

1.4 Multiplexing or multiple access

hta1 p. 16 to 19

Classify the downlink of a mobile tower and the uplink of the phones. Who assigns the channels in multiplexing? Why can CDMA codes not be fully orthogonal in multiple access?

Tower downlink: multiplexing (one source sends several streams; the sender assigns the channels and tells the receivers which is for whom). Phone uplink: multiple access (independent sources). CDMA: the sources are not perfectly synchronized, so the codes cannot be fully orthogonal.

1.5 Duplexing

hta1 p. 20

A 20 Mbit/s radio link uses TDD and gives 3/4 of the time to the downlink. Throughput in each direction? How do Ethernet or USB cables do duplexing? With FDD, what fixes the capacity of each direction?

Downlink 15 Mbit/s, uplink 5 Mbit/s. Cables use a separate wire for each direction. With FDD each direction has its own frequency band, so the split is set by the bands.

1.6 Channel arbitration

hta1 p. 21

Give the slide example for each method: fixed allocation, central entity on request, contention. Which method has collisions?

Fixed: carrier frequencies in cellular mobile. Central entity: time slots in cellular mobile. Contention: Ethernet CSMA, the winner transmits; collisions happen only with contention.

1.7 Communication patterns and addressing

hta1 p. 22 and 23

For unicast, broadcast, multicast and geocast, what kind of address does the communication need?

Unicast: a unique identifier per node. Broadcast: possibly no address (everybody receives). Multicast: a group address, with join and leave. Geocast: geographical coordinates, the nodes need positioning. Unicast and anycast are point-to-point; broadcast, multicast, geocast point-to-multipoint.

1.8 Assigning addresses

hta1 p. 24; hta2 p. 6

Give three ways to assign addresses. In IPv4, what does an address identify? At least how many addresses does a router with three interfaces have?

Manually; one node gives addresses to the others (DHCP); each node chooses its own. An IPv4 address identifies an interface, and one interface can have several: at least 3.

1.9 Centralized and distributed models

hta1 p. 25

Compare the two models and give one example pair from the slides.

Centralized: a central entity coordinates; efficient, but must be installed, trusted, and is vulnerable (needs a fallback). Distributed: entities decide among themselves by contention; less efficient, more fault tolerant. Examples: cellular vs Wi-Fi; HTTP server vs BitTorrent.

1.10 Protocol, stack, SAP, PDU

hta1p p. 2 and 3

Define a communication protocol. What is a Service Access Point (give example methods)? With which layers does a layer talk directly, and with which one virtually?

A set of rules (syntax, semantics, algorithms, state machines) that lets entities exchange information. SAP: the methods to access a protocol's features, e.g. Connect(), Disconnect(), Send(), Receive(); the network API is typically BSD sockets. A layer talks directly only to the layers above and below, and virtually (PDUs) to the same layer on the other end.

1.11 OSI and TCP/IP models

hta1p p. 5 and 6

Why is OSI not used in practice? What does TCP/IP define instead of layer responsibilities? Which OSI layer handles character encoding, compression and encryption?

OSI is too complex and its standard implementation only exists on paper. TCP/IP defines protocols. Presentation layer.

1.12 Encapsulation

hta1p p. 4; hta2 p. 10; hta5 p. 13

1000 octets of application data are sent over TCP and IPv4, both without options. What is the IPv4 Total Length? What may a layer add besides a header?

TCP header 20 octets (Data Offset 5), IPv4 header 20 (IHL 5): $1000 + 20 + 20 = 1040$ octets. A layer may also add a footer.

1.13 Standardization and testing

hta1p p. 7 and 8

In the black box model, what is rigorously specified and what is not? What is a conformance test, and fuzz testing?

Specified: what is visible from outside (interfaces, PDU formats, message sequences). Not specified: implementation details. Conformance test: the black box must answer given inputs with given outputs. Fuzzing: random inputs to try to crash the software.

1.14 Security mechanisms

hta1p p. 12 to 15

What did Diffie-Hellman solve? What does challenge-response protect against? Why is a simple checksum not enough for integrity? What does AAA stand for?

Secure key exchange: agreeing on a shared secret over an insecure channel. Challenge-response (with a secure random number) protects against replay. An attacker can recompute a plain checksum after changing the message: it must include a shared secret, with a secure hash (SHA). AAA: Authentication, Authorization, Access control.

1.15 QoS: reservation, admission, UNI

hta1p p. 16 and 17

What does a client register when negotiating QoS limits? Who accepts or refuses a request? Where is excess traffic dropped, and where are flows identified?

Average bitrate, maximum bitrate, burst length. Call Admission Control. At the network entrance, the User-Network Interface (UNI); flows are identified typically at the network edge.

1.16 Token bucket

hta1p p. 18

A token bucket has $K = 2$ Mbit/s and $B = 1$ Mbit (one token per bit), and is full. The client sends at 6 Mbit/s. How long is the flow unaffected, and how much data passes in that time?

The bucket drains at $6 - 2 = 4$ Mbit/s: empty after $1/4 = 0.25$ s, during which $6 \times 0.25 = 1.5$ Mbit pass. Afterwards messages without tokens are dropped (2 Mbit/s pass).

1.17 Leaky bucket and RED

hta1p p. 19

A leaky bucket outputs at most $K = 2$ Mbit/s and can hold $B = 1$ Mbit. Input: 6 Mbit/s for 0.25 s, then nothing. Is anything dropped, and when has everything left? What does RED do?

1.5 Mbit arrive, 0.5 Mbit leave during the burst: 1 Mbit queued, exactly B, nothing dropped. The queue empties at $0.25+1/2 = 0.75$ s. RED starts dropping with some probability before the bucket is full; TCP scales back on dropped segments.

1.18 Statistical multiplexing

hta1p p. 20

$N = 200$ users, each active with $p = 5\%$. Give the mean, the binomial variance $Np(1-p)$ and σ. With Chebyshev and $k = 2$, which capacity L bounds the blocking probability by 1/4?

Mean 10, variance 9.5, $\sigma \approx 3.08$; $L = 10 + 2 \times 3.08 = 16.16$, so 17 links, $\Pr \leq 1/4$ (Poisson, $\sigma = \sqrt{10} \approx 3.16$, also gives 17). The slide writes $\sigma = Np(1-p)$, the variance: state which one you use. Erlang is only cited.

1.19 Serialization: byte order

hta1p p. 21

Port 8080 is stored on 16 bits. Give the two octets in network order and in memory on an Intel host. Which function converts it? Why avoid floats?

8080 = `0x1F90`: network (big endian) `1F 90`; Intel (little endian) `90 1F`; `htons()`. Floats are not necessarily IEEE 754: avoid or send as text (precision).

1.20 TLV

hta1p p. 21 and 22

Type and Length take one octet each; type 1 = name (ASCII), type 2 = port (16 bits, big endian); the reader knows no other type. Decode `01 03 77 65 62 09 02 AA BB 02 02 1F 90`.

Type 1, length 3: "web". Type 9, length 2: unknown, skipped thanks to its length. Type 2, length 2: `0x1F90` = port 8080. 13 octets in total.

1.21 ASN.1

hta1p p. 22

What is ASN.1, how are its messages serialized, and why do X.509 certificates use DER? Name two other users.

An interface description language (ITU-T X.680); an ASN.1 compiler generates the encoding and decoding code. Serialized in TLV with several encoding rules (BER, CER, DER; PER is packed). DER gives unambiguous decoding. Also SS7, SNMP, 2G to 5G signalling (PER).

1.22 XML tools

hta1p p. 24

What are DTD and XSD, XPath, XSLT, namespaces and SAX used for?

DTD, XSD: schema validation (the document has the required structure). XPath: select elements (e.g. /NTALecture/slides). XSLT: restructure a document (Turing-complete). Namespaces: mix document types in one file. SAX: standard parser API.

1.23 YAML to JSON and XML

hta1p p. 24 to 26

Convert to JSON and to XML (root `<config>`, list items `<port>`). Then give two YAML problems from the course.

```
service: web
ports:
  - 80
  - 443
admin:
  name: Kata
  mail: kata@example.org
```

```
{ "service": "web", "ports": [80, 443],
  "admin": { "name": "Kata",
             "mail": "kata@example.org" } }

<config>
  <service>web</service>
  <ports><port>80</port><port>443</port></ports>
  <admin><name>Kata</name>
    <mail>kata@example.org</mail></admin>
</config>
```

YAML problems: no schema validation; indentation errors; truncation cannot be detected (optional start and end markers).

Session 2: IPv4

2.1 Network layer tasks

hta2 p. 2

Give the three tasks of the network layer and the difference between circuit and packet switching.

Addressing (unique identifiers), routing (find a path between any node pair), forwarding ({DST, SRC, previous hop} → next hop). Circuit switched: a virtual circuit is set up before sending. Packet switched: every packet is routed individually.

2.2 IPv4 service and hourglass model

hta2 p. 3, 4 and 17

What does IPv4 expect from the link layer and offer upwards? In the hourglass model, which layers are private to neighbours, private to endpoints, shared by everyone? Why is IPv4 hard to replace?

Expects: pass the packet to the next hop, no reliability or ordering expectation. Offers: deliver to the destination, no guarantee of reliability or ordering, connectionless, best effort (DiffServ supported). Link layer private to neighbours, upper layers private to endpoints, the network layer must be spoken by everyone. Millions of devices cannot be upgraded at once and a gradual change partitions the network; the reason to replace it is address exhaustion (IPv6).

2.3 Direct delivery, gateway, addressing modes

hta2 p. 5 and 6

Host 10.1.2.3/24 with gateway 10.1.2.1 sends to 10.1.2.77 and to 10.1.3.5. Direct or through the gateway? Can its broadcast reach 10.1.3.0/24?

10.1.2.77 has the same prefix: directly reachable over the link layer. 10.1.3.5 does not: sent to the configured gateway. Broadcast only works within the subnet: no.

2.4 Hop-by-hop and source routing

hta2 p. 7; hta4 p. 6

Compare both, with one advantage and one drawback each, and where source routing is used.

Hop-by-hop: every node decides the next hop with its own topology information, packet by packet; the source needs no topology, but has no control over the path. Source routing: the source discovers the topology and prescribes the path in the message; it controls the path but the network must trust the source. Not used on the Internet; used in ad hoc sensor networks.

2.5 Routing loops and TTL

hta2 p. 8; hta3 p. 4

A packet leaves with TTL 12, crosses R1, R2, R3, then enters a loop R4, R5, R4… Which router drops it, and how many times does it cross the R4-R5 link? Why don't routers just remember the packets they have seen?

Each node decrements the TTL; the 12th router brings it to 0 and drops it: R1 to R3, then R4, R5, R4, R5, R4, R5, R4, R5, R4: dropped by R4, after 8 crossings. It sends ICMP Time Exceeded to the source. Nodes cannot remember seen packets and packets cannot store visited nodes.

2.6 Fragmentation

hta2 p. 9 and 12

A packet with Total Length 2400 (20-octet header) must cross a link with MTU 1006. Give data size, Total Length, Fragment Offset and MF of each fragment. Does IPv4 do path MTU discovery?

Data 2380; per fragment $\lfloor 986/8 \rfloor \times 8 = 984$. Fragments: 984, 984, 412 octets; Total Length 1004, 1004, 432; Offset 0, 123, 246 (units of 8 octets); MF 1, 1, 0; same Identification. Path MTU discovery is not in IPv4.

2.7 IPv4 header fields

hta2 p. 10 to 15

Decode the first 10 octets of a header: `45 00 05 DC 1C 46 40 00 40 11`.

Version 4, IHL 5 (20 octets); DSCP/ECN 0; Total Length `0x05DC` = 1500; Identification `0x1C46` = 7238; flags `010`: DF = 1, MF = 0, offset 0; TTL `0x40` = 64; Protocol `0x11` = 17, UDP.

2.8 IHL, options, checksum, DSCP, ECN

hta2 p. 10, 11, 14, 16

IHL is 7: header length and options length? Why are options rarely used? Why must every router recompute the Header Checksum? What are DSCP and ECN for?

$7 \times 4 = 28$ octets, 8 of options. Options were experimental, are mostly deprecated (RFC 6814) and some routers drop packets carrying them. The checksum covers only the header, including the TTL, which changes at every hop. DSCP: Differentiated Services, only in private networks. ECN: a router signals congestion without dropping (rarely used).

2.9 Classful addressing

hta2p p. 3 and 4

Give the class and say whether it could be assigned to a host before CIDR: 10.200.0.1, 127.0.0.1, 130.50.255.255, 200.1.1.0, 200.1.1.1, 230.1.2.3, 245.0.0.1.

10.200.0.1: A, yes. 127.0.0.1: loopback, no. 130.50.255.255: B broadcast, no. 200.1.1.0: C network address, no. 200.1.1.1: C, yes. 230.1.2.3: D (multicast), no. 245.0.0.1: E (reserved), no. Class B: 16,384 networks of 65,534 hosts.

2.10 Problems of classes, subnetting

hta2p p. 5

An organisation has 2000 computers. How many class C blocks does it need, and which single CIDR prefix length would do? What is the three-level hierarchy of subnetting?

$\lceil 2000/254 \rceil = 8$ class C blocks (8 routing table lines); one /21 ($2^{11} - 2 = 2046$). Subnetting: network prefix, subnet number (e.g. department), host number, inside a class A or B.

2.11 CIDR notation

hta2p p. 6, 7 and 11

Convert 255.255.252.0 to a prefix length and /13 to a netmask. Is 255.255.0.255 valid? How many hosts in a /19?

/22; 255.248.0.0; no (ones then zeros); $2^{13} - 2 = 8190$.

2.12 Special and private addresses

hta2p p. 4 and 8

Which are private, loopback or neither: 172.15.3.1, 172.31.255.1, 192.169.1.1, 10.255.0.1, 127.0.0.53?

172.15.3.1 neither (172.16.0.0/12 is 172.16 to 172.31); 172.31.255.1 private; 192.169.1.1 neither; 10.255.0.1 private; 127.0.0.53 loopback. The slide writes 10.0.0.0/24 and 127.0.0.0/24: it means /8.

2.13 CIDR allocation strategy

hta2p p. 9

How should addresses be allocated for manageable routing tables, and why are IPv4 tables "destined to be a mess forever"?

Large chunks geographically, then subdivided for organisations in the region. CIDR came too late: class B and C blocks were already allocated randomly and class A blocks were broken up. IPv6 started with systematic allocation.

2.14 Mobile IP

hta2p p. 10

Why does moving a device to another network break its connections, and what does Mobile IP do?

The address contains the network prefix, so a new location needs a new address. Mobile IP: an official home address, plus an address in the foreign network; the device stays reachable at its home address.

2.15 Same subnet, dotted mask

hta2p p. 12

Are 10.20.30.40 and 10.20.33.1 on the same subnet with mask 255.255.248.0?

/21, block size 8 in the 3rd octet: 30 gives 10.20.24.0, 33 gives 10.20.32.0. No.

2.16 Longest prefix match

hta2p p. 13

Table: 172.16.0.0/12 if0; 172.20.0.0/14 if1; 172.22.128.0/17 if2; 0.0.0.0/0 if3. Interfaces for 172.23.200.1, 172.22.130.5, 172.31.0.1, 172.32.0.1?

172.23.200.1: /12 and /14 (20 to 23) match, not the /17 (2nd octet 23): if1. 172.22.130.5: if2. 172.31.0.1: if0. 172.32.0.1: default, if3.

2.17 Routing table aggregation

hta2p p. 14

The lines 192.0.4.0/24 to 192.0.8.0/24 (five /24) all go to eth1. Merge them.

4 to 7 form 192.0.4.0/22 (only the last prefix bit differs at each step). 8.0/24 cannot join: 4 to 8 is not an aligned block. Two lines: 192.0.4.0/22 and 192.0.8.0/24.

Session 3: IP protocols and tools

3.1 Data, control, management planes

hta3 p. 2

Place HTTP, RTCP, SNMP, ICMP and OSPF in the planes of the slide. How has the definition of routing changed? Where are ARP and DHCP?

HTTP: data. RTCP: control. SNMP: management. ICMP: control (and management in the old table). OSPF: management in the old definition, control nowadays; management is now only remote configuration. ARP and DHCP sit between layers; their plane is left open by the slide.

3.2 Signalling

hta3 p. 3 and 4

In-band or out-of-band? Why is ICMP in-band for IP but out-of-band for TCP and UDP? How do ITU-T and IETF differ?

In-band: the same channel as user data; out-of-band: a separate channel. ICMP is carried directly in IP (protocol 1), next to user packets, but outside the TCP or UDP connection. ITU-T: a dedicated signalling channel (frequency or time slot); IETF: just IP packets.

3.3 ICMP messages

hta3 p. 4 to 6

Which ICMP type for: TTL reaches 0; datagram to a closed port; DF set but MTU too small; a router knows a better route; ping?

Time Exceeded, 11; Destination Unreachable, 3 (port); Destination Unreachable, 3 (fragmentation needed); Redirect, 5; Echo Request 8 and Echo Reply 0. Router discovery (types 9, 10) is replaced by DHCP.

3.4 ICMP security

hta3 p. 7

Give three abuses of ICMP and what firewalls do.

No authentication: forged Destination Unreachable or Redirect; denial of service (ping flood); ping tunnel; topology discovery of remote networks. Firewalls restrict ICMP (no Echo Reply, Unreachable only from the gateway) or filter it entirely.

3.5 DHCP server, options, tools

hta3 p. 8 to 10; hta3p p. 15

A server hands out 192.168.50.100 to 192.168.50.149 with 1 hour leases. How many clients at most? How are options encoded, and which are typical? Name the usual client and server.

50 addresses. Options are TLV encoded: netmask, default gateway, DNS server, time server, hostname, boot image. The lease is soft state, renewed or it expires. Client dhclient, server Kea (both ISC).

3.6 Several DHCP servers

hta3 p. 10

Two DHCP servers are active on one LAN. What happens during DORA, and why is Request broadcast?

Both may answer the Discover with an Offer; the client picks one. Request is broadcast so that every server learns which offer was chosen.

3.7 URL and URI

hta3 p. 12

Split `https://docs.example.org/net/tcp.html`. Which part does DNS resolve? URL vs URI?

Scheme `https`, location `docs.example.org` (resolved by DNS), path `/net`, file `tcp.html`. A URI is more generic (parts may be missing or added).

3.8 TLDs and FQDN

hta3 p. 13

Classify .hu, .org, .gov and .co.uk. What is an FQDN?

.hu: ccTLD; .org: non-profit; .gov: US government; .co.uk: second level used instead of .com. FQDN: fully specified location in the tree.

3.9 DNS records and lookup

hta3 p. 14 and 15

Which record does a mail server need to find the MTA of example.net? Who holds a subtree? What is a reverse lookup?

MX. Resolvers are responsible for subtrees (e.g. *.tmit.bme.hu); this topology is independent of the network topology. Reverse lookup: from an IP address to an associated name.

3.10 Domain name ownership

hta3 p. 16

Who manages TLDs and root servers, who runs .org, how do you rent a name and check its owner?

ICANN; the Public Interest Registry, created by ISOC; accredited registrars rent the use of names; WHOIS. Free subdomain services also exist.

3.11 DNS transport and security

hta3 p. 17

When is TCP used instead of UDP? Name four weaknesses. Why does DNS over HTTPS bypass the original mechanism?

TCP/53 for longer messages and reliability. No encryption (the site name is visible), no authentication, cache poisoning, look-alike names (other TLD, typos, internationalized letters). DoH sends queries to one provider's servers instead of the resolver tree.

3.12 DNS tools and servers

hta3 p. 18

Name three query tools and two servers.

host, nslookup, dig. BIND 9, Unbound, Microsoft DNS, Dnsmasq.

3.13 Configuration commands

hta3p p. 4, 10 to 14

With iproute2: give eth0 address 172.16.5.20/24, bring it up, add a route to 10.40.0.0/16 via 172.16.5.1. Old equivalents? What does `netstat -pnaut` list? Windows equivalent of ifconfig?

`ip addr add 172.16.5.20/24 dev eth0; ip link set dev eth0 up; ip route add 10.40.0.0/16 via 172.16.5.1 dev eth0`. Old: `ifconfig`, `route`. netstat: open connections with process id, numeric, all states, UDP and TCP. Windows: ipconfig.

3.14 Permanent settings

hta3p p. 16 and 17

Write `/etc/network/interfaces` for eth0 by DHCP and eth1 static 172.16.5.20/24, gateway 172.16.5.1. Where is the DNS server configured? Which tool has its own DHCP client?

```
auto eth0
auto eth1
iface eth0 inet dhcp
iface eth1 inet static
    address 172.16.5.20/24
    gateway 172.16.5.1
```

`/etc/resolv.conf`, managed by dhclient. systemd-networkd (NetworkManager uses nmcli; netplan sits on top of both).

3.15 Ping and traceroute

hta3p p. 18 to 20

Can ping succeed while web traffic fails? Which probes can traceroute send?

Yes, the slides warn about it (e.g. other traffic filtered). Traceroute: UDP to a random port by default, ICMP Echo Request with `-I`, TCP SYN with `-T`.

3.16 Capture tools

hta3p p. 21 to 24

What do tcpdump, Wireshark and tshark do? What is promiscuous mode?

tcpdump records and decodes traffic, can save to file (tcpreplay). Wireshark (formerly Ethereal) is the graphical equivalent; tshark its console counterpart. Promiscuous mode captures traffic addressed to other nodes.

3.17 Mininet

hta3p p. 25; hta4p p. 15

What is Mininet and what does its Python API do?

A sandbox of lightweight virtual machines connected by virtual links and switches; the API builds a network and runs it with a command line (terminals, Wireshark on the nodes).

Session 4: routing

4.1 Routing, forwarding, RIB, FIB

hta4 p. 2 and 3; hta4p p. 2

Define forwarding and routing. What can the RIB contain that the FIB does not? What makes a Linux machine a router?

Forwarding: choose a next hop with the topology information; routing: discover the topology (static or dynamic). RIB: topology discovery, may hold discovery method, timeouts; FIB: optimized for lookup speed. Any node that forwards is a router: `/proc/sys/net/ipv4/ip_forward`.

4.2 Router capacity

hta4 p. 3

A 10 Gbit/s interface: packets per second with 1500-octet packets, and with 64-octet packets?

$10^{10}/12{,}000 \approx 833{,}333$ pps; $10^{10}/512 \approx 19.53$ Mpps. Packet rate depends on processing, hence a separate capacity figure.

4.3 Optimal path and redundancy

hta4 p. 5

List four optimality criteria. With 1+1 redundancy over two disjoint paths each losing a packet with probability 1% independently, what is the probability of losing both copies? Why disjoint?

Fewest hops, shortest distance, lowest latency, highest bandwidth, lowest cost, highest reliability. $0.01^2 = 0.01\%$. A shared link would lose both copies at once. 1:1 keeps a backup path instead.

4.4 Flat and hierarchical routing

hta4 p. 6

Compare flat and hierarchical models.

Flat: one domain, only for small networks. Hierarchical: domains; intra-domain routing for nodes inside, inter-domain routing for the domain gateways.

4.5 Distance vector convergence

hta4 p. 7; hta4p p. 4 and 5

8 routers form a ring (hop count). After how many rounds does every router know every destination? What does an advertisement contain?

Diameter 4: 4 rounds ("diameter 10 converges in 10 rounds"). Pairs (destination, distance); the next hop is stored but not advertised.

4.6 RIP hop limit

hta4 p. 10; hta4p p. 6

A RIP network has a diameter of 17 hops. Consequence? Why weren't more fields added to the advertisements to fix count to infinity?

Destinations 16 hops or more away are unreachable (16 means infinity, limit 15). The fixes (split horizon, poisoned reverse) avoid changing the protocol.

4.7 Link state database

hta4p p. 7 and 8

Links W-X 3, W-Y 1, X-Y 1, X-Z 4, Y-Z 6. Write the link state database and the shortest distances from W. Give the four steps of link state routing.

W: X/3, Y/1. X: W/3, Y/1, Z/4. Y: W/1, X/1, Z/6. Z: X/4, Y/6. From W: Y 1, X 2 (via Y), Z 6 (W-Y-X-Z). Steps: discover neighbours (Hello), exchange LS (LSA) with known nodes, repeat with newly discovered nodes, run Dijkstra.

4.8 Algorithm complexity

hta4p p. 3

$|V| = 1000$, $|E| = 5000$: compare $|E| + |V| \log |V|$ (base 2) with $|V|^2$. Which algorithm for LS, which for DV, and why?

About 14,966 vs 1,000,000. LS uses the faster Dijkstra; Bellman-Ford can be distributed, so DV uses it.

4.9 RIP versions

hta4 p. 10

What changed in RIPv2 and RIPng? Metric and update interval of RIP?

RIPv2: CIDR, faster convergence (shorter interval), authentication. RIPng: IPv6, authentication by IPsec. Hop count, 30 s.

4.10 OSPF

hta4 p. 11

What are areas and the backbone? Is 0.0.0.0 an address here? Why a Designated Router? Metric?

LSAs stay inside an area; the backbone area 0.0.0.0 connects all others; the 32-bit area ID is not an IP address. On non point-to-point links (Ethernet) a DR (and BDR) represents the segment. Metric: path cost, typically bandwidth relative to a reference.

4.11 IS-IS

hta4 p. 12

Give three differences with OSPF.

Not tied to IP (any layer 3, also SPB); no central backbone area; easier to extend; used by ISPs.

4.12 Autonomous systems

hta4 p. 13

What is an AS? Pros and cons? IGP and EGP examples?

The network of one provider or company. Easier operation, scalability; difficult on a global scale. IGP: RIP, OSPF, IS-IS. EGP: BGP.

4.13 Peering and transit

hta4 p. 14

ISP1 peers with ISP2, ISP2 peers with ISP3. Can ISP1 reach ISP3 through ISP2?

No: peering is pairwise and not transitive. A transit provider (higher level AS, paid by contract) connects lower level ASes.

4.14 BGP

hta4 p. 15

Number of iBGP sessions for a full mesh of 12 routers? With 3 route reflectors in full mesh, each serving 3 clients that peer only with their reflector? Why is BGP routing "policy routing"?

$12 \times 11/2 = 66$. Full mesh between the 3 RRs (3) plus one session per client (9): 12. Path cost includes money and trust between ASes, besides bandwidth or latency; path vector: whole paths are exchanged.

4.15 Implementations and demo

hta4 p. 16; hta4p p. 15 to 17

What does a routing daemon do? Name two open-source ones. In the demo, why does a ping to an address on a non-adjacent link fail before BIRD starts?

It runs in the background and fills the FIB. Quagga, FRRouting, XORP, BIRD. Only directly connected /24 networks are known until OSPF or RIP adds the routes.

Session 5: transport layer and TCP

5.1 Transport and session in the IP model

hta5 p. 2

Who handles keepalive and multiplexing in TCP/IP compared with OSI?

OSI session: connect, disconnect, keepalive; OSI transport: reliable segments, multiplexing. In TCP/IP keepalive and multiplexing are the application's responsibility.

5.2 Identifiers

hta5 p. 3

Size and scope of MAC address, IP address and port? How many port numbers?

MAC 48 bits, interface locally; IP 32 or 128 bits, interface globally; port 16 bits, a connection of an application: 65,536 values.

5.3 Port allocation and 5-tuple

hta5 p. 4

A browser opens two connections to the same server on port 443. What distinguishes them? Which ports are well-known? Which protocol fixes both ports?

The ephemeral source port, so the 5-tuple {IP 1, IP 2, protocol, port 1, port 2} differs. 0 to 1023 (IANA, superuser on Unix). DHCP: server 67, client 68.

5.4 Flow and congestion control

hta5 p. 5

Who is protected by each, and how is congestion detected?

Flow control: the receiver (not infinite processing power). Congestion control: the network. Detected by increased latency or packet drop.

5.5 Datagrams and streams

hta5 p. 6; hta5p p. 20

An application writes three 100-octet messages to a TCP socket. Will the receiver read three 100-octet messages? And with UDP?

Not necessarily: TCP is a stream, reads do not match writes; read until the message end (e.g. `</html>`). UDP: one message per segment, delivered immediately.

5.6 UDP

hta5 p. 7

UDP carries 100 octets: Length field and IPv4 Total Length? What does UDP add to IP?

$100 + 8 = 108$; $108 + 20 = 128$. Ports (no reliability, flow or congestion control).

5.7 TCP, UDP and in between

hta5 p. 8, 20 to 23

Place TCP, UDP, DCCP and SCTP. Why are QUIC and μTP built over UDP?

TCP: connection, stream, reliable, ordered, flow and congestion control, no broadcast or multicast. UDP: connectionless, datagram, unreliable. DCCP: UDP with reliable setup, ACK without retransmission, congestion notification. SCTP: datagrams, ordered or not, multihoming (telecom). Other sockets need special privileges.

5.8 Connection start and end

hta5 p. 9 and 10

Who may close a connection, with which segments? Why are initial sequence numbers random? Other attacks besides SYN flood?

Either party: FIN, FIN/ACK, ACK. Random ISNs make guessing harder. Connection hijacking and reset (only cited).

5.9 Acknowledgement and sliding window

hta5 p. 11; hta5p p. 2

Window of 4 segments; ACK 3 has arrived. Which segments may be sent? And after ACK 5?

3 to 6; then 5 to 8: the window advances when its first segment is acknowledged.

5.10 TCP header

hta5 p. 13 to 16

Data Offset is 8: header and options length? What do PSH and URG mean? Largest window with window scaling?

32 octets, 12 of options (TLV: window scale, SACK, timestamp). PSH: push data to the application; URG: important data until the Urgent Pointer. $65{,}535 \times 2^{14} \approx 2^{30}$: 1 GB.

5.11 NAT trade-offs

hta5 p. 18 and 19

A gateway can use 28,000 ephemeral ports: how many simultaneous connections to one server port? Two benefits and two problems of NAT?

28,000 (one external port per connection). Benefits: more computers than addresses, hidden network. Problems: limited ports, peer-to-peer and H.323, signed endpoint addresses.

5.12 Flow control, Silly Window Syndrome

hta5p p. 3

What happens when adv_wnd = 0? Causes and fixes of the Silly Window Syndrome?

The sender stops, then sends a small segment to get a new window. SWS: a fast sender fills the buffer, small windows follow; or small application writes. Fixes: Nagle's algorithm (collect writes), advertise only when the buffer is half empty.

5.13 Bottleneck

hta5p p. 4

A path has links of 1 Gbit/s, 100 Mbit/s, 20 Mbit/s, 1 Gbit/s. Bottleneck? Why do bottlenecks move?

20 Mbit/s, not the first link. Load changes (daily cycle, competing flows) create new bottlenecks.

5.14 Congestion collapse and goodput

hta5p p. 5 and 6

Load 100 Mbit/s, of which 30 Mbit/s are duplicates or lost. Goodput? Where should the load stay? Why not reserve bandwidth for every flow?

70 Mbit/s (70%). Left of the knee point, close to it. Reservation is an NP-complete packing problem, does not scale and delays connection setup.

5.15 Effective window

hta5p p. 7

MSS 1460 octets, cwnd 10 MSS, adv_wnd 8760 octets. Effective window?

$\min(14{,}600, 8760) = 8760$ octets = 6 MSS: flow control limits here.

5.16 Slow start and short connections

hta5p p. 8

cwnd starts at 1, ssthresh is large, no loss. How many round trips to send 60 segments? Why does HTTP suffer?

1, 2, 4, 8, 16, 32: 63 segments after 6 round trips. Short connections stay in slow start.

5.17 Tahoe and Reno

hta5p p. 9 to 15

cwnd 1, ssthresh 8, slow start capped at ssthresh, CA +1; a loss is detected at the end of round 8. Give cwnd for rounds 1 to 12 with Tahoe (timeout) and Reno (3 duplicate ACKs), and the segments sent in rounds 9 to 12.

Tahoe: 1, 2, 4, 8, 9, 10, 11, 12, 1, 2, 4, 6 (ssthresh 6). Reno: 1, 2, 4, 8, 9, 10, 11, 12, 6, 7, 8, 9. Rounds 9 to 12: 13 vs 30 segments, the larger area under the Reno curve.

5.18 Long Fat Network

hta5p p. 16

Satellite link: 10 Mbit/s, RTT 600 ms. Bandwidth-delay product, throughput without window scaling, minimum shift?

$10^7 \times 0.6/8 = 750{,}000$ octets. $65{,}535 \times 8/0.6 \approx 0.874$ Mbit/s. 2^3 gives 524,280, too small; 2^4 gives 1,048,560: shift 4.

5.19 Links with high loss rate

hta5p p. 17

Why does Reno perform badly on Wi-Fi, and which solutions do the slides give?

Reno takes loss for congestion and scales back. Delay-based window, fast retransmit, SACK (New Reno defers shrinking with SACK).

5.20 TCP variants

hta5p p. 18 and 19

Match: New Reno, Vegas, Compound, CUBIC. Which variant is the default today?

New Reno: retransmit on the first duplicate ACK, keep cwnd. Vegas: delay-based, loses bandwidth to Reno. Compound: loss and delay windows (Vista). CUBIC: cwnd a cubic function of the time since the last congestion, mostly RTT independent; replaced BIC in Linux, New Reno in OS X, Compound in Windows 10.

5.21 Programmers and operators

hta5p p. 20 and 21

Why reuse connections? What is `SO_REUSEPORT` for? Why is TCP hard for operators?

Setup is slow (3-way and TLS handshakes): HTTP/1.1. Rebinding a port that is still lingering after close. TCP traffic is bursty, fills the available bandwidth, and variants are not fair to each other (Compound pushed out Reno).

Session 6: Web and e-mail

6.1 World Wide Web

hta6 p. 2 and 3

What is hypertext? Directory vs search engine? What does robots.txt do? Who standardizes?

Documents linked by hyperlinks, identified by URLs, no dead link detection. Directory: catalog in categories; search engine: crawlers. robots.txt disables indexing. W3C.

6.2 Browsers

hta6 p. 4

What is a browser engine? Which engines are still developed, and why is Blink's dominance a problem?

HTTP client, HTML renderer, JavaScript runtime. Blink (Chrome, Edge, Opera, Brave), Gecko (Firefox), WebKit (Safari). Google dictates the web: interoperability and free access suffer.

6.3 Servers and proxies

hta6 p. 5

Name two web servers and four uses of a web proxy.

IIS, Apache, nginx, Lighttpd. Load balancing, caching, filtering, protocol translation.

6.4 HTML

hta6 p. 6

What is XHTML, what superseded it, and what describes formatting separately?

HTML that is valid XML; HTML5; CSS.

6.5 JavaScript

hta6 p. 7

What are the DOM and AJAX?

DOM: API to interact with the HTML document. AJAX: load content asynchronously (XMLHttpRequest), nowadays with JSON.

6.6 HTTP methods and response codes

hta6 p. 8

Which method to fetch a page, to send a search term, to get only metadata? Meaning of 200, 301, 403, 404, 500, 502, 504?

GET, POST, HEAD. 200 OK; 301 Moved Permanently; 403 Forbidden; 404 Not Found (client errors); 500 Internal Server Error; 502 Bad Gateway; 504 Gateway Timeout (server errors).

6.7 User-Agent and Referrer

hta6 p. 9

Why do browsers lie in the User-Agent string? What is the Referrer for?

Servers tailor content to buggy browsers; after fixes, browsers pretend to be another one. Referrer: the page that linked to the document (how people find a site).

6.8 HTTP versions

hta6 p. 10

A page has 1 HTML file and 19 images on one server. TCP handshakes with one connection per object vs HTTP/1.1 reuse? What do HTTP/2 and HTTP/3 add?

20 vs 1. HTTP/2 (SPDY): header compression, request multiplexing. HTTP/3: QUIC instead of TCP.

6.9 State and cookies

hta6 p. 11

Why can't a response with a cookie be cached, and the workaround? What is a third-party cookie?

It is user specific; serve static content from a separate domain without cookies. A cookie set on behalf of another domain, used for tracking; fingerprinting tracks even without cookies.

6.10 Symmetric and asymmetric cryptography

hta6 p. 12

What does Diffie-Hellman not protect against? Encrypt for Bob, sign as Alice: which keys? Why sign a hash?

Man in the middle. Encrypt with Bob's public key; sign with Alice's private key. Asymmetric crypto is slow: sign a secure hash (SHA-256). RSA was the first working method.

6.11 Publishing public keys

hta6 p. 13

Web of trust vs PKI. Three criticisms of PKI?

WoT: trust whom your trusted friends trust (PGP). PKI: CAs sign site keys in X.509 certificates. Criticisms: foreign CAs trusted by default; CAs selling wiretapping; how to know the CAs' keys.

6.12 SSL and TLS

hta6 p. 14

Relation between SSL and TLS, deprecated versions, layer, StartTLS?

After SSL 3.0 comes TLS 1.0; older than TLS 1.2 is deprecated. Application layer protocol, after the TCP handshake. StartTLS upgrades an insecure connection.

6.13 QUIC drawbacks

hta6 p. 15

Give three limits of QUIC from the slides.

Congestion control unspecified; mostly used by Google, poor server support; mandatory encryption is bad for caching proxies; TCP to QUIC redirect not clearly defined.

6.14 E-mail origins

hta6p p. 3

What did `username@computer` mean originally, and now?

A login on a Unix mainframe; now the address is not tied to a login, only a mailbox.

6.15 Agents

hta6p p. 4

Role of MTA, MDA, MUA, Message Submission Agent, Message Retrieval Agent?

MTA relays (SMTP). MDA: a relay that stores, holds the mailbox. MUA: the user's client, with submission and retrieval parts.

6.16 Message format

hta6p p. 5

Write the header of a message from Anna Kis to two recipients, then where the body starts. Give a MIME type.

```
From: "Anna Kis" <anna@example.org>
To: bob@example.net, eva@example.net
Subject: Lab report
```

then an empty line and the 7-bit ASCII body. MIME type: text/plain, application/octet-stream.

6.17 Mail storage

hta6p p. 6

Compare mbox and maildir.

mbox: all mails in one file, each starts with From:. maildir: one file per mail (hashed name), subdirectories, each with new, cur, tmp.

6.18 SMTP

hta6p p. 7

Command sequence, transport, ports, response codes?

MAIL, RCPT, DATA; TCP, text based; 25 between servers, 465 (formerly 587) for submission, TLS; codes organized like HTTP.

6.19 POP3 and IMAP commands

hta6p p. 8 and 9

Name the commands and how IMAP handles the Sent folder.

POP3: RETR, DELE; IMAP: FETCH; both answer OK. The client uploads the message to Sent after the SMTP send.

6.20 Mail software

hta6p p. 10 to 12

How is webmail usually implemented? Name an MTA. What is special about Exchange?

As an IMAP client. Sendmail, Exim, Postfix. Proprietary protocol (also POP3/IMAP), calendar.

6.21 Client authentication

hta6p p. 13

Three factors? When is PLAIN acceptable? How does CRAM work and what does it prevent?

Know, have, are. Only inside TLS. Server sends a random number, client returns hash(password, number), server compares: protects against replay.

6.22 OpenID, OAuth, TOTP

hta6p p. 14

What does each provide?

OpenID: single login at an identity provider. OAuth 2.0: identity and a token to act for the user (e.g. IMAP to Gmail). TOTP: 6-digit code from a shared secret and the time (2FA).

6.23 PGP and S/MIME

hta6p p. 15

What does PGP use instead of CAs? What is S/MIME for?

Web of trust (asymmetric keys, OpenPGP, GnuPG). Signed or encrypted attachments; hard with webmail.

6.24 Fighting spam

hta6p p. 16 to 18

Two MTA acceptance policies and what MUA filters look at.

Accept only from authenticated MUAs, only from respected MTAs (plus DKIM, SPF, DMARC). Keywords, suspicious header patterns, AI pattern recognition.

6.25 Phishing

hta6p p. 19

What is phishing and why is full protection impossible?

Social engineering to reveal information or install malware; attacks are sophisticated and one accidental click is enough.